Received: 24 December 2016 | Revised: 3 January 2017 | Accepted: 26 February 2017

DOI: 10.1111/crj.12621

REVIEW ARTICLE

WILEY

Tuberculosis associated chronic obstructive pulmonary disease

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Abstract

Objectives: Reviewed the epidemiology, clinical characteristics, mechanisms, and treatment of tuberculosis associated chronic obstructive pulmonary disease.

Data source: We searched PubMed, EMBASE, and the CINAHL from inception to June 2016. We used the following search terms: Tuberculosis, COPD, Tuberculosis associated COPD, and so forth. All types of study were chosen.

Results and Conclusion: Chronic obstructive pulmonary disease (COPD) and tuberculosis are significant public health problems, particularly in developing countries. Although, smoking is the conventional risk factor for COPD, nonsmoking related risk factors such as biomass fuel exposure, childhood lower-respiratory tract infections, chronic asthma, outdoor air pollution, and prior history of pulmonary tuberculosis have become important risk factors of COPD, particularly in developing countries. Past history of tuberculosis as a risk factor of chronic airflow obstruction has been reported in several studies. It may develop during the course of tuberculosis or after completion of tuberculosis treatment. Developing countries with large burden of tuberculosis can contribute significantly to the burden of chronic airflow obstruction. Prompt diagnosis and treatment of tuberculosis should be emphasized to lessen the future burden of chronic airflow obstruction.

KEYWORDS

chronic obstructive pulmonary disease, smoking, TOPD, tuberculosis, vitamin D deficiency

1 | INTRODUCTION

The global initiative for chronic obstructive lung disease (GOLD)[1] defined chronic obstructive pulmonary disease (COPD) as "a common preventable and treatable disease, characterized by persistent airflow limitation that is usually progressive and associated with an enhanced chronic inflammatory response in the airways and the lung to noxious particles or gases." COPD is a major public health problem worldwide due to its high prevalence, morbidity, and mortality.[2,3] It also poses a significant socioeconomic burden.[4] The global prevalence of spirometry–defined COPD is 11.7%.[2] The recent World Health Organization (WHO) report revealed that in the year 2012, more than 3 million people died of COPD, and more than 90% of these deaths occur in developing countries.[3] The mortality due to COPD is also showing an increasing trend. COPD was the fourth leading cause of death in 1990 and at present, it is the third leading cause of death globally.[5] Tuberculosis is a major killer of mankind. According to World Health Organization (WHO) report, in the year 2014, 9.6 million people developed active tuberculosis and 1.5 million died due to tuberculosis. It is the leading infectious cause of death globally.[6] About 58% of world's tuberculosis cases occurs in South-East Asia and India contributes to 23% of the global tuberculosis burden.[6] Developing countries are facing the dual epidemics of communicable diseases like tuberculosis, and human immunodeficiency virus infection (HIV) infection, and noncommunicable diseases like smoking and COPD.[7] The convergence of these epidemics may increase the susceptibility to diseases. Optimal management of these conditions requires thorough assessment of all the interacting epidemics.

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Smoking is the conventional risk factor for COPD but definitely not the only risk factor. Nonsmoking related risk factors such as biomass fuel smoke exposure, childhood lower-respiratory tract infections, chronic asthma, outdoor air pollution, crop and animal farming, exposure to various dust, chemicals and pollutants, and treated case of pulmonary tuberculosis are also important risk factors in disease causation, particularly in developing countries.[8,9] Past history of pulmonary tuberculosis has recently emerged as a risk factor for later development of COPD. If this relationship is correct, it would pose a significant burden on healthcare considering the fact that prevalence of both the conditions are high in developing countries. Post-tuberculous development of COPD is not a new thought; it has been mentioned in the literature in the past also.[10–14] However, this issue has been studied in details only recently. Allwood et al. named the condition of tuberculosis-associated obstructive pulmonary disease as TOPD.[15] However; it is still a matter of debate whether chronic airflow obstruction and TOPD are different or same entities. This commentary will review the epidemiological and mechanistic links between tuberculosis and COPD/chronic airflow obstruction.

1.1 | Tuberculosis and airflow obstruction

Patients of pulmonary tuberculosis may develop airflow obstruction either during the active phase or post-treatment phase of the disease. The prevalence of airflow obstruction in pulmonary tuberculosis is variable depending on the nature of study, definition of airflow obstruction used, and geographical locations. The relation between past history of tuberculosis and later development of chronic airflow obstruction has been reported even in 1950s and 1960s. However, the major drawback in early studies was lack of adequate controls; therefore; the confounding effect of smoking on airflow obstruction was not adjusted.[16]

1.1.1 | Epidemiological studies

Anno and Tomashefski in one of the early study reported impairment of respiratory function in pulmonary tuberculosis patients. There was an increase in the residual volume (RV), residual volume/total lung capacity ratio and a reduction in the maximal breathing capacity in a selected group of tuberculosis patients.[10] Gaensler et al.[17] in a study done in 1959, reported evidence of airflow obstruction in 61% of 1533 tuberculosis patients. In a cross-sectional study, snider et al.[18] evaluated 1403 patients of pulmonary tuberculosis admitted to a Chicago Municipal tuberculosis Sanatorium from 1964 to 1966 and reported airflow obstruction [defined by forced expiratory volume in one second and forced vital capacity (FEV1:FVC) ratio $<70\%$] in 23% patients. The authors noticed no significant association between smoking and airflow obstruction despite a very high prevalence of smoking in the study patients. There was no association between duration of active disease prior to presentation with airflow obstruction. Airflow obstruction was seen more commonly in patients with history of morning phlegm. Several studies in recent years had further reinforced the earlier findings of positive association between tuberculosis and development of chronic airflow obstruction. Lam et al.[19] in a population-based study involving Guangzhou Biobank Cohort reported past history of tuberculosis as an independent predictor of airflow obstruction with odd ratio of 1.37 after adjusting sex, age, and smoking exposure. However, in this study, pre-bronchodilator FEV_1:FVC ratio was used instead of conventional post-bronchodilator FEV_1/FVC ratio to define airflow obstruction. Lee et al.[20] in a cross-sectional population-based study based on data from Second Korea National Health and Nutrition Examination Survey 2001, analyzed the effects of previous tuberculosis on the risk of obstructive lung disease development. Total 294 subjects had evidence of previous tuberculosis on chest radiograph. After adjustment for sex, age, and smoking history, radiological evidence of previous tuberculosis was independently associated with airflow obstruction with the odds ratios of 2.56. The risk was present even in patients with minimal radiographic changes also. The multicenter population-based PLATINO study[21] also evaluated the relationship between past history of tuberculosis and development of airflow obstruction in five Latin American countries. Subjects aged ≥ 40 years underwent pre- and post-bronchodilator spirometry. The overall prevalence of airflow obstruction was 30.7% among those with a history of tuberculosis, compared with 13.9% among those without a history of tuberculosis with the odd ratio of 2.33 after adjustment for confounders like age, sex, schooling, ethnicity, smoking, exposure to dust and smoke, respiratory morbidity in childhood and current morbidity. The risk is higher in males than in females with adjusted odds ratio of 3.99 and 1.71, respectively. The association between tuberculosis and COPD was even stronger in never smokers. History of prior tuberculosis also indicates more severe form of COPD. The burden of obstructive lung disease (BOLD) study had also shown past history of tuberculosis as a risk factor for developing airflow obstruction in later life with the adjusted odd ratio of 2.51.[22] Two recent systemic reviews also confirmed this association.[23,24]

Allwood et al.[23] in a systematic review assessed the relation between pulmonary tuberculosis and the development of chronic airflow obstruction. Total 19 studies including 1 case series, 3 case-control studies, 4 cohort studies, 8 single-center cross-sectional studies, and 3 multicenter cross-sectional studies were eligible for final analysis. The authors confirmed a positive association between a past history of tuberculosis and the presence of chronic airflow obstruction. This relationship was independent of cigarette smoking and

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biomass fuel exposure. Byrne et al.[24] in another systematic review and meta-analysis found that the association between prior history of tuberculosis and the presence of COPD was strongest in never smokers and younger people (<40 years). Development of chronic airflow obstruction also depends on the geographical area. It is higher in high tuberculosis incidence countries. The odds of chronic airflow obstruction development were more than three times in countries with a reported tuberculosis incidence of more than 100/100 000 population/year (adjusted OR 3.13 for Korean adults;[20] 4.9 and 6.6 for South African males and females, respectively,[25] and 6.31 for the Philippines[26]).

1.1.2 | Clinical characteristics of patients with TOPD

There are few studies that have compared the clinical features of patients of TOPD with conventional smoking-related COPD. TOPD involves people with young age (<40 years)[24] as tuberculosis-associated lung damages occur earlier whereas smoking-associated lung damages occurs slowly and later.[23] Lee et al.[27] in a case-control study evaluated lung function parameters of 21 TOPD patients and compared it with COPD patients matched by age-, sex-, and FEV_1% predicted. Hemoptysis was more common in TOPD compared to COPD patients as patients with TOPD may develop bronchiectasis. No significant differences were noted for dyspnea, cough, exacerbations, and hospitalizations between these two groups. Patients of TOPD also had significantly lower FVC and post-bronchodilator FEV_1 value compared to those of COPD patients. Positive bronchodilator response was significantly lower in TOPD than in COPD patients, indicating irreversible nature of airflow obstruction. Airway resistance was also higher in patients with TOPD, although; it was not statistically significant. Seo et al.[28] studied the clinical features of TOPD and COPD patients admitted in the intensive care unit (ICU) and observed a significantly higher frequencies of tuberculous pneumonia and tracheostomies in patients of TOPD compared to COPD patients. Park et al.[29] retrospectively reviewed the data of 38 patients of tuberculous destroyed lungs from South Korea admitted to the ICU and mechanically ventilated. Majority was males and 52% were nonsmokers. Pulmonary function tests reports available for 21 patients in the preceding 12 months showed very severe obstruction with a mean FEV_1 of 0.77 L (29.3% predicted), mean FVC of 1.52 L (41% predicted) and a mean FEV_1/FVC ratio of 55.1%, respectively. However, due to the limitations in study designs, the impact of smoking could not be ruled out. Airflow limitation in patients with tuberculous destroyed lungs is an independent risk factor for acute exacerbation.[30] Kim et al.[30] found significantly higher frequencies of acute exacerbation in patients with airflow limitation compared to those without airflow limitation (89.1 vs. 67.7%, respectively; $P = .009$). The annual decline of FEV_1 was lower in patients with airflow limitation group. It was 2 mL in patients with airflow limitation and 36 mL in those without ($P < .001$). The rate of FEV_1 decline depends on the baseline FEV_1 value. Larger the baseline FEV_1 value, greater is the rate of FEV_1 decline. Lower baseline FEV_1 value in the airflow limitation group can explain this finding. Rhee et al.[31] in a retrospective study analyzed the clinical characteristics of 595 patients with tuberculous destroyed lungs and compared the patients with and without airflow limitation. Mean FVC, FEV_1 and FEV_1/FVC ratio, and bronchodilator response were $2.06 \pm .03$ l ($61.26\% \pm 0.79$), $1.16 \pm .02$ l ($49.05\% \pm 0.84$), $58.03\% \pm 0.70$, and $5.70\% \pm 0.34$, respectively. They also found significant correlation between the number of lobes involved with FVC and FEV_1 and exacerbation frequencies. Therefore; patients with extensive parenchymal involvement are at high risk of exacerbations and should be given special attention. Follow up data of pulmonary functions results were also obtained to see the trend in FEV_1 over time. The FEV_1 value in patients with tuberculous destroyed lungs showed a decreasing trend like COPD with a mean annual decrement of 38.24 ± 7.98 mL. The change in FEV_1 on multivariable regression analysis showed correlation with age, initial FEV_1(%) and number of exacerbations. Bronchodilator response was an important difference between patients of tuberculous destroyed lungs and COPD. It was significantly lower in tuberculous destroyed lungs group than in COPD. It signifies the presence of anatomical obstruction of airways caused by destruction of lung tissues and bronchial stenosis. The authors also showed significant improvement in FEV_1 with the use of long-acting muscarinic antagonists (LAMA), long-acting beta-2 agonists plus inhaled corticosteroids (LABA + ICS). Gunen et al.[32] divided COPD patients hospitalized due to exacerbation into two groups: based on presence or absence of radiological old tuberculosis scars. Past tuberculosis scars was characterized by earlier development and first progression of COPD. However, mortality rate was similar between the two groups. There was total 598 COPD patients and 93(15.8%) had old tuberculosis scars. COPD patients with a history of tuberculosis were younger ($P = .002$). The first COPD diagnosis in patients with old tuberculosis scars was also made five year earlier compared to patients without tuberculosis scars (58.6 ± 12.3 years vs. 63.2 ± 11.2 years, P value <.001).

1.1.3 | Time course of changes in lung function parameters in patients with pulmonary tuberculosis

Pulmonary tuberculosis patients usually develop maximum loss of lung function within six months of the diagnosis of

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tuberculosis and it stabilizes 18 months after completion of treatment.[33,34] Hnizdo et al.[35] observed a fall in FEV_1 values of 56.8 mL per month with the lowest FEV_1 value (326 mL) seen at approximately 6 months post-tuberculosis treatment and it stabilized 7 to 12 months later. Vargha et al.[36] measured vital capacity (VC), FEV_1, and RV in patients with tuberculosis discharged in 1958/59. Total 40 patients had obstructive pattern based on the FEV_1/VC% ratio. Reassessment was done in 1974. The change in FEV_1 was 35.3 mL/year. However, the impact of other noxious environmental factors could not be ruled out during such a long period of follow-up.[34] Rhea et al. reported a mean annual decrement in the FEV_1 value of 38.24 ± 7.98 mL.[31]

Frequency and severity of airflow obstruction in pulmonary tuberculosis depends on the number of episodes of tuberculosis. Hnizdo et al.[35] studied this relationship in 27 660 South African Gold miners. Frequency of chronic airflow obstruction is directly proportional to the number of episodes of tuberculosis. The prevalence of chronic airflow obstruction after one episode of tuberculosis was 18.4%, whereas the prevalence after two and ≥ 3 episodes were 27.1% and 35.2%, respectively. Structural damages of the lungs increase with the increasing number of tuberculosis episodes and they persist in a large number of patients despite anti-tubercular chemotherapy. The reported fall in FEV_1 was also highest in this study, probably related to compounding effects of other factors like smoking and dust exposures; adjustment for these was not done in this study. Increase number of tuberculosis episodes also accelerates the loss of FEV_1 and the average loss after one, two, and three or more tuberculosis episodes were 153 mL, 326 mL, and 410 mL, respectively. Plit et al.[33] prospectively evaluated the impact of anti-tubercular chemotherapy on lung function of newly diagnosed pulmonary tuberculosis patients. Anti-tubercular chemotherapy resulted in improved lung function in 54% of patients. However, residual airflow limitation or restrictive pattern was seen in 28% and 24% of patients, respectively. The residual lung functions impairment depends on both pre- and post-treatment radiological extent of the disease. This study suggested that lung functions impairment may develop even after successful treatment of tuberculosis.

Various predictors of impaired pulmonary functions have been reported in pulmonary tuberculosis patients and these include smear-positive disease, extensive pulmonary involvement prior to anti-tubercular treatment, reduced radiographic improvement post-treatment, and prolonged duration of treatment.[34] Delay in initiating anti-tuberculosis treatment is also an independent risk factor of COPD. In a population-based cohort study done in Taiwan, Lee et al. evaluated the impact of pulmonary tuberculosis, delayed initiation and nonadherence to anti-tubercular treatment on the future risk of COPD.[37] Prior history of pulmonary tuberculosis was found to be an independent risk factor for developing COPD (hazard ratio 2.054 [1.768–2.387]). The risk persisted for at least six years after tuberculosis diagnosis. Delayed initiation of anti-tubercular treatment was an independent risk factor as it prolonged the duration and increased the severity of airway inflammation. It results in accelerated rate of lung destruction and subsequent loss of lung function. Therefore, early diagnosis and prompt initiation of anti-tuberculosis treatment is essential for controlling both tuberculosis and COPD epidemics.

Patients with chest radiological changes incurred by tuberculosis should be a candidate for screening COPD. Hwang et al.[38] in a population-based survey evaluated the relation between radiologic changes by tuberculosis with airflow obstruction. Airflow obstruction was defined by GOLD criteria. The prevalence of airflow obstruction in subjects with radiological changes was 26.3%. It was significantly higher than patients without any radiologic changes. The unadjusted odds ratio for airflow obstruction based on radiologic change was 3.788 (95% CI: 2.544–5.642) and when adjusted for smoking, it was over 3.12. Patients with radiological changes were in higher GOLD stages. It can be explained by greater lung function loss in tuberculosis patients with radiological changes. Ross et al.[39] had shown accelerated loss of lung functions among South African Gold miners following pulmonary tuberculosis. The mean annual loss of FEV_1 was 40.3 mL/year after adjusting age, height, baseline lung function, silicosis, years of employment, smoking, and other respiratory diagnoses. Interestingly; airflow obstruction occurred mainly in patients with minimal radiologic changes, indicating that the mechanism of airflow obstruction is chronic airway inflammation rather than airway fibrosis seen in advanced radiologic involvement.

Pulmonary tuberculosis is also a stronger risk factor for chronic airflow obstruction than smoking. The prevalence of COPD in Colombia (PREPOCOL) study evaluated the prevalence and risk factors of COPD in five Colombian cities situated at different altitudes. A strong association between a history of tuberculosis and the development of airflow obstruction was seen (odds ratio of 2.94) and it was greater than that with smoking.[40] Similarly, Ehrlich et al.[25] in a nationwide survey of 13 826 adults in South Africa, found that a history of pulmonary tuberculosis was a stronger predictor of COPD than tobacco smoking or biomass fuel smoke exposure. Ngahane et al.[41] studied the prevalence and predictors of lung functions impairment in a tuberculosis reference clinic in Cameroon. They reported lung function impairment in 45.4% of patients. Duration of symptoms and fibrotic radiologic pattern was independent risk factors for lung function impairment on multivariate analysis. Treated case of pulmonary tuberculosis patients may develop distal airflow obstruction defined by forced expiratory flow at 25–75% of forced vital capacity ($FEF_{25–75\%}$)

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<65% and a ratio $FEV_1/FVC \geq 0.70$. Pefura-Yone et al.[42] reported distal airflow obstruction in 62.9% of successfully treated patients of pulmonary tuberculosis. Distal airflow obstruction was independently associated with persisting chronic pulmonary signs.

The risk of later COPD development is not specific for *Mycobacteria tuberculosis*. It has been reported even with non-tuberculosis mycobacterium (NTM) species. Yeh et al.[43] in a Taiwanese cohort of 3005 NTM patients showed that NTM patients have 3.08-fold increased risk of developing COPD compared with the non-NTM cohort. Patients of NTM with abnormal pulmonary function initially showed improvement after macrolides therapy.[44] Early diagnosis and treatment of NTM disease should be an important goal to prevent future COPD development.

1.1.4 | Indian studies

There are few Indian studies evaluating the relationship between past history of pulmonary tuberculosis and COPD. Study done in 100 fully treated pulmonary tuberculosis patients in a tertiary care teaching hospital in India, Brashier et al.[45] reported a 46% prevalence of airflow obstruction and the prevalence increased with the duration after treatment completion. Airflow obstruction was mild ($FEV_1 > 60\%$) in 75%, moderate (FEV_1 40–59%) in 10%, and severe ($FEV_1 < 40\%$) in 15% patients. In the Gothi et al.[46] series, obliterative bronchiolitis was the cause of chronic airflow obstruction in 13% of patients and 78% of obliterative bronchiolitis was post-tubercular. Verma et al.[47] reported a much lower prevalence of airflow obstruction. In fully treated pulmonary tuberculosis patients, obstructive spirometric pattern was seen in 15(16.3%) patients and majority had irreversible airflow obstruction.

2 | COMMON RISK FACTORS

There are some common risk factors for both the conditions such as smoking, low socioeconomic status (SES), biomass fuel exposure and vitamin D deficiency.[48,49] Smoking is a conventional and well-established risk factor for COPD.[50] In addition; there is considerable evidence now available to incriminate smoking as a risk factor of tuberculosis also. There are three systematic reviews and meta-analyses that have implicated active smoking as a risk factor for tuberculosis infection, active tuberculosis disease, and tuberculosis mortality.[51–53] The impact of active smoking is more on tuberculosis disease than on tuberculosis infection. Other effects of active smoking include risk of recurrence of tuberculosis and tuberculosis mortality. Active smoking can cause recurrent tuberculosis after satisfactory completion of treatment for a previous tuberculosis episode. Yen et al.[54] had shown that smoking more than 10 cigarettes a day increased the risk of recurrence of tuberculosis by 2-fold compared to that of never/former smokers. Active smoking also increases risk of lost to follow-up, greater severity of disease, drug resistant tuberculosis, slower smear conversion, and mortality.[55] Because of the dual epidemic of tuberculosis and smoking present in many developing countries, the impact of smoking on tuberculosis would be enormous. Number of tobacco users in India is an enormous 275 million.[56] Therefore; public health policy must include smoking cessation as a part of national tuberculosis control programme. The international union against tuberculosis and lung disease has already endorsed this view.[57] Lin et al.[58] had shown that the complete cessation of tobacco and solid-fuel use by 2033 would reduce the tuberculosis incidence by 14–52% if 80% DOTS coverage is maintained. The mechanistic link between smoking and tuberculosis is not clear but may involve the immunologic and oxidative pathway.[7] Most of the immunological abnormalities developed due to smoking are reversible within six weeks of stopping smoking.[59]

2.1 | Biomass fuel exposure

In developing countries, biomass fuel exposure is the most important nonsmoking related cause of COPD.[9] Worldwide; number of people exposed to biomass fuel smoke is much higher than those exposed to smoking.[59] There are several studies from India, Nepal and Brazil that reported biomass fuel smoke exposure as an independent risk factor for tuberculosis.[60–63] Misra et al.[60] found that the odds ratio of developing active tuberculosis was 3.56 in persons living in households that mainly use biomass fuel for cooking. This risk is significant as a large number of people use biomass fuel for heating or cooking in developing countries.

2.2 | Low socio-economic status

Low SES is a risk factor both COPD and tuberculosis. SES is a composite measure of several indices such as income, education, occupation, house conditions, location of residence, and crowding index.[64] Low SES is also responsible for poor health related quality of life in COPD patients.[65] Kanervisto et al.[66] in a population-based study had shown basic educational level as an independent risk factor for COPD (odd ratio of 1.8). In a case-control study involving 250 consecutive tuberculosis patients, Gupta et al.[67] found-poor SES as a risk factor for tuberculosis. In multivariate logistic regression analysis, following factors were found to be significantly and independently associated with a higher risk of tuberculosis; age, level of education, crowding, type of housing, water supply, and number of consumer articles in the household. There are several risk factors that a person

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with low SES is exposed such as malnutrition, indoor air pollution, alcohol, crowded and less ventilated places, unhealthy cooking practice that may increase the risk of tuberculosis.[68]

2.3 | Vitamin D Deficiency

Vitamin D deficiency is a potential risk factor for both tuberculosis and COPD. Nnoaham et al.[69] in a systematic review and meta-analysis evaluated the relation between low serum vitamin D levels and risk of active tuberculosis. Seven observational studies published between 1980 and July 2006 was included for analysis. Higher risk of active tuberculosis was seen in patients with low serum Vitamin D level. Zeng et al.[70] in a meta-analysis identified the serum Vitamin D ranges associated with increase tuberculosis risk. They reported a statistically significant association between serum Vitamin D level of ≤ 25 nmol/L with risk of tuberculosis. Nursyam et al.[71] in a randomized clinical trial studied the impact of Vitamin D supplementation on tuberculosis outcome. Compared to placebo, vitamin D supplementation resulted in higher rate of sputum conversion and radiological improvement. Vitamin D plays an important role in host immune defense against *Mycobacterium tuberculosis* via innate and adaptive immune system.[72,73]Vitamin D also has a role to play in COPD initiation, COPD pathogenesis, COPD exacerbations and development of musculoskeletal comorbidities of COPD.[74] Early life event has been recognized as a risk factor for COPD and any event that impairs lung growth at early life may lead to an increased COPD risk.[75] Vitamin D is required for lung development and its deficiency has the potential to initiate chronic lung diseases at early age.[76]

2.4 | Diabetes mellitus

The interaction of diabetes with tuberculosis is bi-directional. Jeon et al.[77] in a systematic review and meta-analysis of 13 observational studies evaluated the relation between diabetes and tuberculosis. There is a 3-fold higher risk of developing active tuberculosis in diabetics compared with people who are nondiabetics. The risk is higher in insulin-dependent diabetes mellitus (IDDM) than in noninsulin dependent diabetes mellitus (NIDDM). Olmos et al.[78] in a longitudinal-retrospective study reported the 10-year actuarial probability of developing tuberculosis of 24% in IDDM and 4·8% in NIDDM. There is also increased risk of progression to active disease in diabetics patients.[79] Baker et al.[80] in a systematic review showed that tuberculosis patients with diabetes are at a higher risk of failure, relapse, and death while on anti-tubercular treatment. Diabetes and COPD also interact in bi-directional way. The progression and prognosis of COPD is worsened by comorbid diabetes. It does so by several mechanisms: direct effects of hyperglycemia on lung physiology, inflammation and/or susceptibility to bacterial infection. The systemic inflammation and use of corticosteroids may also increase the risk of developing type-2 diabetes.[81]

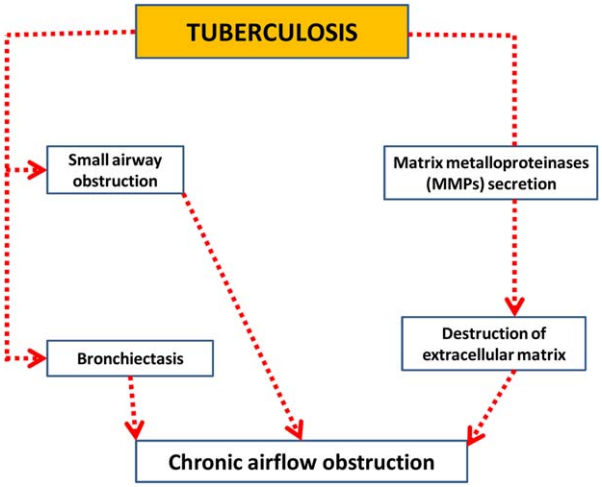


FIGURE 1 is showing the mechanisms of airflow obstruction due to tuberculosis

3 | MECHANISMS OF AIRFLOW OBSTRUCTION DUE TO TUBERCULOSIS

The exact mechanism of CAO in post-tuberculosis patients is not clear. Following mechanisms have been proposed for the development of tuberculosis-associated CAO. These include bronchiectasis, bronchiolar narrowing and bronchiolitis obliterans and accelerated emphysematous changes.[23] Figure 1 is showing the mechanisms of airflow obstruction due to tuberculosis.

3.1 | Small airway involvement

Small airways are non-cartilaginous airways with an internal diameter <2 mm.[82] Tuberculosis lesions may involve the small airways that results in airflow obstruction. Im et al. reported centrilobular nodules, "tree-in-bud" appearance and poorly defined nodules in 95% of individuals with pulmonary tuberculosis.[83] These lesions disappear by 5 months after initiation of chemotherapy. In tuberculosis endemic countries, tuberculosis is an important cause of obliterative bronchiolitis. In the Gothi et al.[46] series, 92% cases of obliterative bronchiolitis were post infectious, and 78% of them being post-tubercular. Characteristics radiological feature of bronchiolitis is gas trapping characterized on computed tomography (CT) scan as areas of reduced attenuation on expiratory CT scans. This radiologic feature may persist after tuberculosis treatment although the endobronchial and parenchymal changes undergo resolution.[84] Jeong et al.[85] also reported small airway involvement in patients with NTM

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infection. Allwood et al. evaluated patients with healed pulmonary tuberculosis and airflow obstruction by dynamic quantitative CT lung imaging and spirometry, plethysmography and diffusion capacity. Patients with chronic airflow obstruction with definite previous TB had higher gas trapping, fibrosis and emphysema score than subjects with no previous tuberculosis. The diffusion capacity was also significantly lower in patients with definite previous TB.[15]

3.2 | Bronchiectasis

Post-tuberculous bronchiectasis is common in developing countries particularly in patients who had a history of recurring tuberculosis.[86] Palwatwichai et al.[87] in a prospective study from Thailand found a history of tuberculosis in 32% of 50 patients with bronchiectasis, making it the most common underlying etiology of bronchiectasis. Natarajan et al.[88] in a prospective study from Western India reported allergic bronchopulmonary aspergillosis and post-tuberculous sequelae as the commonest causes seen in 23% each. Bronchiectasis develops due to endobronchial obstruction or peribronchial fibrosis or obstruction by enlarged lymph nodes and is an important cause of chronic airflow obstruction.[89] Bronchiectasis within areas of scarring is called tractional bronchiectasis. Tuberculosis patients may also develop bronchostenosis and features of airway obstruction. Chae et al.[90] reported atelectasis (84%) and emphysema with bronchiectasis (89%) as the most common CT findings in patients with tubercular destroyed lungs.

3.3 | Accelerated parenchymal destruction

Lung parenchymal inflammation in tuberculosis may lead to destruction of the pulmonary extra-cellular matrix (ECM), a feature similar to that of COPD.[91] Matrix metalloproteinases (MMPs) are a family of calcium-dependent zinc-containing endopeptidases[92] that mediates tissue remodeling in tuberculosis.[93] MMPs are involved in the pathogenesis of both tuberculosis and COPD by degrading components of ECM, the scaffold of the alveolar wall. Different MMPs have different role in mycobacterial infection. MMP-9 helps in production of stable granuloma, thereby containing the infection. However, reactivation of latent TB leads to MMP-1 secretion. MMP-1 causes alveolar destruction and is responsible for cavitation in tuberculosis.[94] Finlay et al.[95] found alveolar macrophages as a significant source of MMPs in the emphysematous lung. They found significant quantity of MMPs particularly MMP-9 in BAL fluid of COPD patients compared to normal healthy volunteers. Similarly, Imai et al. found increased expression MMP-1 in the lung parenchyma of COPD patients compared to normal controls. It is the type II pneumocyte which makes MMP-1 and takes part in lung destruction.[96] Therefore, MMPs are a common mediator that may connect both the disorders. However, we need more studies to clearly elucidate MMPs role in the pathogenesis of tuberculosis and COPD. Singh et al.[97] studied the impact of various antimycobacterial agents on MMPs levels in tuberculosis patients. Levels of MMPs like MMP-1, −2,-3, −7, −8, and −9 were elevated in the broncho-alveolar lavage fluid of tuberculosis patients and showed correlation with tuberculosis-associated tissues damages. Various antimycobacterial agents for example rifampin, moxifloxacin, and azithromycin showed an immunomodulatory effect by decreasing the epithelial cell gene expression and secretions of MMPs. We need further study to explore whether antitubercular therapy can prevent future COPD development by inhibiting MMPs expression in the airway epithelial cells.

4 | IMPACT OF COPD ON TUBERCULOSIS

Patients of COPD are also at increased risk of developing tuberculosis. Inghammar et al.[98] studied the impact of COPD on tuberculosis incidence and mortality. A total of 115 867 patients of aged more than 40 years and discharged from Swedish hospital with a diagnosis of COPD were evaluated for tuberculosis risk. There was a 3-fold higher risk of developing tuberculosis in COPD patients than controls and the incidence of tuberculosis was inversely related to FEV_1 value. In addition; the risk of death from all causes was 2-fold higher within first year after the tuberculosis diagnosis in COPD patients with active tuberculosis compared to general population control subjects with tuberculosis. Lee et al.[99] studied the risk factors for pulmonary tuberculosis in COPD patients. Among 23 594 COPD cases and 47 188 non-COPD control subjects, cox regression analysis revealed COPD as independent risk factors for tuberculosis with the hazard ratio of 2.47. COPD patients who developed tuberculosis received more oral corticosteroids and oral β-agonists. Inhaled corticosteroid use has also been related to a higher risk of tuberculosis and the risk is more with the use of high dose of inhaled corticosteroids equivalent to fluticasone 1000 mg/d or more.[100] Therefore, in tuberculosis endemic countries, it should be mandatory to rule out tuberculosis infection/disease before high dose inhaled corticosteroid is used. Corticosteroids increase the risk of tuberculosis by causing immune suppression.[101] Shang et al.[102] studied the immunological response to *M. tuberculosis* in cigarette smoke exposed mice. The lung and splenic macrophages and dendritic cells released lower levels IL-12 and TNF-α in cigarette smoke-exposed mice. Therefore, cigarette smoke exposure alters the protective immune response to *M. tuberculosis*. The immune impairment in patients with pulmonary tuberculosis and COPD combined is much more

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than each of the individual disease.[103] Tang et al.[103] measured various cytokines in the peripheral blood of 152 cases of pulmonary tuberculosis-COPD combined, 150 cases of patients with pulmonary tuberculosis, 157 cases of patients with COPD, and 50 cases of healthy volunteers. The tuberculosis-COPD combined patients had significantly lower percentage of CD4+ T cells than that in the non-COPD tuberculosis patients. The tuberculosis patients had significantly higher percentage of CD8+ T cells than that in control group. Cytokines like sIL-2R, IL-6, TNF-α, IFN-γ levels were significantly higher in COPD patients with tuberculosis than those in COPD patients without tuberculosis. High levels of these cytokines by producing an exuberant inflammatory response may cause progression of tuberculosis in COPD patients. COPD patients itself develop dysfunction of alveolar macrophages independent of steroids which poses an additional risk of tuberculosis.[104,105] Macrophages are one of the key cells in tuberculosis and also play a role in wound healing and resolution.[106] They may cause remodeling of the airways. Mycobacteria within lungs may lead to uncontrolled activation of wound-healing macrophages, resulting in remodeling of the airways and development of chronic airflow obstruction.[107]

5 | TREATMENT OF TOPD

Treatment of TOPD is no difference than treatment of COPD alone. Yum et al.[108] studied the efficacy of inhaled tiotropium bromide for 2 months period on 29 patients with tuberculous destroyed lungs. There was a significant improvement in FEV_1 and FVC value over baseline with tiotropium therapy. Lee et al.[27] found that patients of chronic airflow obstruction due to tuberculous destroyed lung presenting with wheezing showed better bronchodilator response. Despite all these results, it is noteworthy to mention that treatment of COPD is not variable depending on etiological factors of origin.

6 | FUTURE PERSPECTIVE

There are several arenas which are still unexplored at the present moment. First of all; we need to have large population-based case control study with patients of tuberculosis and asymptomatic normal population to prove the causality between tuberculosis and COPD. We need to explore the differences between TOPD and COPD in relation to clinical, physiological, radiological, airway inflammation, and response to treatment. We need to know whether there are any differences in terms of development of airflow obstruction between children and the adults afflicted with pulmonary tuberculosis. Unraveling these areas in future will definitely provide better ideas about the condition TOPD.

7 | CONCLUSION

Past history of tuberculosis is an important risk factor of COPD and tuberculosis is also common in COPD patients. Development of chronic airflow obstruction may be related to common risk factors or tuberculosis associated lung damages. Both COPD and tuberculosis are major health problem in developing countries. Early diagnosis and institution of adequate treatment of tuberculosis should be emphasized to reduce the future burden of COPD. Emphasis should also be given to avoid common risk factors like smoking and biomass fuel smoke exposure to prevent development of both tuberculosis and COPD in future.

CONFLICT OF INTEREST

The authors have stated explicitly that there are no conflicts of interest in connection with this article.

AUTHOR CONTRIBUTIONS

Dr. Malay Sarkar, Dr. Irappa Madabhavi, Dr. Kushal Kumar, Dr. Srinivasa: collecting data. Dr. Irappa Madabhavi, Dr. Kushal Kumar: English correction. Dr. Malay Sarkar: write up.

ETHICS

No human or animal studies were done so no ethical clearance is required.

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How to cite this article: Sarkar M, Srinivasa, Madabhavi I, Kumar K. Tuberculosis associated chronic obstructive pulmonary disease. *Clin Respir J.* 2017;11:285–295. https://doi.org/10.1111/crj.12621

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